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JEE MAIN PHYSICS DPP: ERROR ANALYSIS & PROPAGATION

Topic: Systematic & Random Errors, Relative/Percentage Errors, Propagation | Time: 45 Min | Max Marks:

80

Marking Scheme: +4, -1 | Designed & Curated by: Syed Mazhar Ali

Q1. A physical quantity P is given by the relation $P = \frac{A^3 B^{1/2}}{C D^2}$. The percentage errors in the measurement of A, B, C , and D are 1%, 2%, 3%, and 2% respectively. The maximum permissible percentage error in P is:

- (a) 11%
- (b) 14%
- (c) 9%
- (d) 7%

Q2. In an experiment to determine g using a simple pendulum, length $L = 100.0$ cm is measured with a scale of least count 1 mm, and time for 100 oscillations is $T = 200.0$ s measured with a stopwatch of resolution 0.1 s. The maximum percentage error in g is:

- (a) 0.2%
- (b) 0.1%
- (c) 0.05%
- (d) 0.3%

Q3. The resistance $R = V/I$, where $V = (100 \pm 5)$ V and $I = (10 \pm 0.2)$ A. The percentage error in the value of R is:

- (a) 5%
- (b) 2%
- (c) 7%
- (d) 3%

Q4. Which of the following statements regarding systematic errors is **incorrect**?

- (a) They tend to be in one direction, either positive or negative.
- (b) They can be minimized by taking the arithmetic mean of multiple readings.
- (c) Instrumental and zero errors are systematic errors.
- (d) They can be eliminated by calibration.

Q5. The density of a solid sphere is determined by measuring its mass m and diameter d . If percentage errors in m and d are 1.5% and 1% respectively, the maximum percentage error in density is:

- (a) 2.5%
- (b) 3.5%
- (c) 4.5%
- (d) 5.5%

Q6. Two resistors $R_1 = (100 \pm 3) \Omega$ and $R_2 = (200 \pm 4) \Omega$ are connected in series. The equivalent resistance with absolute error limit is:

- (a) $(300 \pm 7) \Omega$
- (b) $(300 \pm 1) \Omega$
- (c) $(300 \pm 3.5) \Omega$
- (d) $(300 \pm 5) \Omega$

Q7. Two resistors $R_1 = (6.0 \pm 0.2) \Omega$ and $R_2 = (3.0 \pm 0.1) \Omega$ are connected in parallel. The equivalent resistance R_p along with its absolute uncertainty is:

- (a) $(2.0 \pm 0.3) \Omega$
- (b) $(2.0 \pm 0.07) \Omega$
- (c) $(2.0 \pm 0.1) \Omega$
- (d) $(2.0 \pm 0.05) \Omega$

Q8. The initial and final temperatures of a liquid are measured as $\theta_1 = (20.4 \pm 0.2)^\circ\text{C}$ and $\theta_2 = (50.6 \pm 0.3)^\circ\text{C}$. The rise in temperature is:

- (a) $(30.2 \pm 0.1)^\circ\text{C}$
- (b) $(30.2 \pm 0.5)^\circ\text{C}$
- (c) $(30.2 \pm 0.25)^\circ\text{C}$
- (d) $(30.2 \pm 0.3)^\circ\text{C}$

Q9. If random errors in an experiment follow a Gaussian distribution, increasing observations by a factor of 16 reduces random error by a factor of:

- (a) 16
- (b) 4
- (c) 2
- (d) 8

Q10. The voltage across a wire is $V = (50 \pm 2)$ V and current is $I = (5 \pm 0.1)$ A. Power dissipated $P = VI$ with error limits is:

- (a) (250 ± 10) W
- (b) (250 ± 15) W
- (c) (250 ± 2.1) W
- (d) (250 ± 5) W

Q11. In Searle's method for Young's modulus $Y = \frac{4MgL}{\pi d^2 l}$, which quantity must be measured with maximum precision to minimize error in Y ?

- (a) Load mass M
- (b) Length of wire L
- (c) Diameter of wire d
- (d) Elongation of wire l

- Q12.** A student measures focal length f of a convex lens using $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$. If $u = -20$ cm and $v = +30$ cm with $\Delta u = \Delta v = 0.1$ cm, the maximum error in focal length Δf is:
- (a) 0.052 cm
(b) 0.065 cm
(c) 0.083 cm
(d) 0.020 cm
- Q13.** The heat generated in a resistor is $H = I^2 R t$. If percentage errors in I , R , and t are 2%, 1%, and 1% respectively, the maximum percentage error in H is:
- (a) 4%
(b) 6%
(c) 8%
(d) 5%
- Q14.** A plate measured with a meter scale has length 10.5 cm and width 4.0 cm. If least count is 0.1 cm, the relative error in area is:
- (a) 0.035
(b) 0.025
(c) 0.010
(d) 0.045
- Q15.** For a simple pendulum $T = 2\pi\sqrt{l/g}$, if $\frac{\Delta l}{l} = 0.02$ and $\frac{\Delta T}{T} = 0.03$, the fractional error in g is:
- (a) 0.08
(b) 0.05
(c) 0.04
(d) 0.01
- Q16.** Refractive index of glass is measured as $\mu = d_1/d_2$, where real depth $d_1 = (4.00 \pm 0.02)$ cm and apparent depth $d_2 = (2.50 \pm 0.02)$ cm. The percentage error in μ is:
- (a) 0.8%
(b) 1.3%
(c) 1.5%
(d) 0.5%
- Q17.** The radius of a sphere is measured to be (2.1 ± 0.1) cm. The percentage error in its surface area is:
- (a) 4.76%
(b) 9.52%
(c) 2.38%
(d) 7.14%
- Q18.** A quantity z is given by $z = x - y$. If $x = (10.0 \pm 0.1)$ and $y = (8.0 \pm 0.2)$, the percentage error in z is:
- (a) 3%
(b) 10%
(c) 15%
(d) 5%
- Q19.** In a specific gravity bottle experiment, $m_1 = (20.0 \pm 0.1)$ g (empty), $m_2 = (40.0 \pm 0.1)$ g (with water), and $m_3 = (36.0 \pm 0.1)$ g (with liquid). The relative density of liquid is:
- (a) 0.80 ± 0.018
(b) 0.80 ± 0.010
(c) 0.80 ± 0.025
(d) 0.80 ± 0.005
- Q20.** The speed of sound is $v = \sqrt{\gamma P / \rho}$. If fractional errors in P and ρ are α and β respectively, the maximum fractional error in v is:
- (a) $\alpha + \beta$
(b) $\frac{1}{2}(\alpha + \beta)$
(c) $\frac{1}{2}(\alpha - \beta)$
(d) $2(\alpha + \beta)$

ANSWER KEY & STEP-BY-STEP SOLUTIONS

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Quick Answer Key

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(a)	6	(a)	11	(c)	16	(b)
2	(a)	7	(b)	12	(a)	17	(b)
3	(c)	8	(b)	13	(b)	18	(c)
4	(b)	9	(b)	14	(a)	19	(a)
5	(c)	10	(b)	15	(a)	20	(b)

Detailed Solutions

Sol 1. Correct Option: (a)

$$\text{For } P = \frac{A^3 B^{1/2}}{C D^2}:$$

$$\frac{\Delta P}{P} = 3 \frac{\Delta A}{A} + \frac{1}{2} \frac{\Delta B}{B} + \frac{\Delta C}{C} + 2 \frac{\Delta D}{D}$$

$$\begin{aligned} \% \text{ error} &= 3(1\%) + \frac{1}{2}(2\%) + 3\% + 2(2\%) \\ &= 3\% + 1\% + 3\% + 4\% = 11\% \end{aligned}$$

Sol 2. Correct Option: (a)

$$T = t/n \Rightarrow \frac{\Delta T}{T} = \frac{\Delta t}{t}$$

$$\text{Since } g = \frac{4\pi^2 L}{T^2}:$$

$$\frac{\Delta g}{g} = \frac{\Delta L}{L} + 2 \frac{\Delta t}{t} = \frac{0.1}{100.0} + 2 \left(\frac{0.1}{200.0} \right)$$

$$\frac{\Delta g}{g} = 0.001 + 0.001 = 0.002 \Rightarrow 0.2\%$$

Sol 3. Correct Option: (c)

$$R = \frac{V}{I} \Rightarrow \frac{\Delta R}{R} = \frac{\Delta V}{V} + \frac{\Delta I}{I}$$

$$\frac{\Delta R}{R} \times 100 = \left(\frac{5}{100} + \frac{0.2}{10} \right) \times 100 = 5\% + 2\% = 7\%$$

Sol 4. Correct Option: (b)

Systematic errors are unidirectional and reproducible. Arithmetic averaging of multiple readings only reduces **random errors**.

Sol 5. Correct Option: (c)

$$\rho = \frac{m}{V} = \frac{6m}{\pi d^3} \Rightarrow \frac{\Delta \rho}{\rho} = \frac{\Delta m}{m} + 3 \frac{\Delta d}{d}$$

$$\frac{\Delta \rho}{\rho} = 1.5\% + 3(1\%) = 4.5\%$$

Sol 6. Correct Option: (a)

$$\text{In series: } R_s = R_1 + R_2 = 100 + 200 = 300 \Omega.$$

Absolute error adds directly:

$$\Delta R_s = \Delta R_1 + \Delta R_2 = 3 + 4 = 7 \Omega$$

$$\text{Thus, } R_s = (300 \pm 7) \Omega.$$

Sol 7. Correct Option: (b)

$$R_p = \frac{R_1 R_2}{R_1 + R_2} = \frac{6.0 \times 3.0}{6.0 + 3.0} = 2.0 \Omega.$$

$$\text{Differentiating } \frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}:$$

$$\Delta R_p = R_p^2 \left(\frac{\Delta R_1}{R_1^2} + \frac{\Delta R_2}{R_2^2} \right)$$

$$\Delta R_p = (2.0)^2 \left(\frac{0.2}{36} + \frac{0.1}{9} \right) = 4 \left(\frac{0.6}{36} \right) \approx 0.07 \Omega$$

$$\text{Result: } (2.0 \pm 0.07) \Omega.$$

Sol 8. Correct Option: (b)

$$\Delta \theta = \theta_2 - \theta_1 = 50.6 - 20.4 = 30.2^\circ \text{C}.$$

In difference, absolute errors add:

$$\delta(\Delta \theta) = 0.2 + 0.3 = 0.5^\circ \text{C}$$

$$\text{Result: } (30.2 \pm 0.5)^\circ \text{C}.$$

Sol 9. Correct Option: (b)

$$\text{Random error } \sigma_{\text{mean}} = \frac{\sigma}{\sqrt{n}}.$$

$$\text{When } n \rightarrow 16n, \text{ error reduces by } \sqrt{16} = 4.$$

Sol 10. Correct Option: (b)

$$\text{Power } P = VI = 50 \times 5 = 250 \text{ W}.$$

$$\frac{\Delta P}{P} = \frac{\Delta V}{V} + \frac{\Delta I}{I} = \frac{2}{50} + \frac{0.1}{5} = 0.06$$

$$\Delta P = 0.06 \times 250 = 15 \text{ W}.$$

$$\text{Result: } (250 \pm 15) \text{ W}.$$

Sol 11. Correct Option: (c)

$$\text{In } Y = \frac{4MgL}{\pi d^2 l}, \frac{\Delta Y}{Y} \propto 2 \left(\frac{\Delta d}{d} \right).$$

Since wire diameter d is squared and very small in magnitude, it must be measured with highest precision.

Sol 12. Correct Option: (a)

$$f = \frac{uv}{u-v} = \frac{(-20)(30)}{-50} = 12 \text{ cm}.$$

$$\text{From } \frac{1}{f} = \frac{1}{v} - \frac{1}{u}:$$

$$\Delta f = f^2 \left(\frac{\Delta v}{v^2} + \frac{\Delta u}{u^2} \right)$$

$$\Delta f = 144 \left(\frac{0.1}{900} + \frac{0.1}{400} \right)$$

$$= 144(0.000361) \approx 0.052 \text{ cm}$$

Sol 13. Correct Option: (b)

$$H = I^2 R t \Rightarrow \frac{\Delta H}{H} = 2 \frac{\Delta I}{I} + \frac{\Delta R}{R} + \frac{\Delta t}{t}.$$

$$\% \text{ error} = 2(2\%) + 1\% + 1\% = 6\%$$

Sol 14. Correct Option: (a)

$$\frac{\Delta A}{A} = \frac{\Delta l}{l} + \frac{\Delta b}{b} = \frac{0.1}{10.5} + \frac{0.1}{4.0} \approx 0.035.$$

Sol 15. Correct Option: (a)

$$g = \frac{4\pi^2 l}{T^2} \Rightarrow \frac{\Delta g}{g} = \frac{\Delta l}{l} + 2 \frac{\Delta T}{T}.$$

$$\frac{\Delta g}{g} = 0.02 + 2(0.03) = 0.08$$

Sol 16. Correct Option: (b)

$$\mu = \frac{d_1}{d_2} \Rightarrow \frac{\Delta \mu}{\mu} = \frac{\Delta d_1}{d_1} + \frac{\Delta d_2}{d_2}.$$

$$\% \text{ error} = \left(\frac{0.02}{4.00} + \frac{0.02}{2.50} \right) \times 100 = 1.3\%$$

Sol 17. Correct Option: (b)

$$S = 4\pi r^2 \Rightarrow \frac{\Delta S}{S} = 2 \frac{\Delta r}{r}.$$

$$\% \text{ error} = 2 \left(\frac{0.1}{2.1} \right) \times 100 \approx 9.52\%$$

Sol 18. Correct Option: (c)

$$z = x - y = 10.0 - 8.0 = 2.0.$$

$$\Delta z = \Delta x + \Delta y = 0.1 + 0.2 = 0.3.$$

$$\% \text{ error} = \frac{0.3}{2.0} \times 100 = 15\%$$

Sol 19. Correct Option: (a)

$$m_w = 40.0 - 20.0 = 20.0 \text{ g}, \Delta m_w = 0.2 \text{ g}.$$

$$m_l = 36.0 - 20.0 = 16.0 \text{ g}, \Delta m_l = 0.2 \text{ g}.$$

$$\text{RD} = \frac{16.0}{20.0} = 0.80.$$

$$\frac{\Delta \text{RD}}{\text{RD}} = \frac{0.2}{16.0} + \frac{0.2}{20.0} = 0.0225$$

$$\Delta \text{RD} = 0.0225 \times 0.80 = 0.018.$$

$$\text{Result: } 0.80 \pm 0.018.$$

Sol 20. Correct Option: (b)

$$v = \sqrt{\gamma P / \rho} \Rightarrow \frac{\Delta v}{v} = \frac{1}{2} \frac{\Delta P}{P} + \frac{1}{2} \frac{\Delta \rho}{\rho} =$$

$$\frac{1}{2}(\alpha + \beta).$$

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